MODULE 25 · DOMAIN 1 - AGENTS AND WORKFLOWS

The Claude Agent SDK and Hosting

What the Agent SDK gives you, custom tools, hooks for deterministic actions, permissions in order, sessions, when a custom loop is justified, and where agents run: self-hosted or Managed Agents.

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Buy a reliable engine; do not cast your own. **The SDK is the engine. You build the car around it.**

What the Agent SDK gives you

- **Claude Code's loop and tools,** programmable in Python and TypeScript.
- **The loop runs for you:** tool dispatch, history, automatic compaction near the limit.
- **Built-in tools:** Read, Edit, Write, Glob, Grep, Bash, WebSearch, WebFetch, Agent, Skill and more.
- **Also built in:** subagents, hooks, permissions, sessions, MCP servers, Skills.
- **Limits default to none:** set max turns and a spend budget yourself.

> ON THE EXAM
>
> The default system prompt is minimal unless you set one. It is not Claude Code's own prompt.

An agent with the SDK

```
options = ClaudeAgentOptions(
    system_prompt=AGENCY_OPS_PROMPT,
    mcp_servers={"property": PROPERTY_MCP, "copperlet": copperlet_tools},
    allowed_tools=["mcp__property__property_get", "mcp__copperlet__payments_lookup"],
    permission_mode="dontAsk",                    # unlisted tools are refused
    hooks={"PreToolUse": [HookMatcher(matcher="^mcp__", hooks=[audit_and_scope])]},
    max_turns=25, max_budget_usd=2.00,
)
async for message in query(prompt=case_brief(case), options=options):
    handle(message)                               # stream progress; final result last
```

4-5 A locked-down agent: only listed tools, nothing else asked or allowed.

6-7 A hook on every MCP call, and hard limits the loop cannot exceed.

THE TAKEAWAY

Custom tools are in-process MCP servers: @tool functions wrapped with create_sdk_mcp_server.

Hooks: deterministic actions

- **Events:** PreToolUse, PostToolUse, UserPromptSubmit, Stop, SubagentStart, PreCompact and more.
- **Matchers on tool names:** Write|Edit, or ^mcp__ for every MCP tool.
- **A PreToolUse hook can allow, deny, ask, or rewrite the input.**
- **Deny wins:** deny beats defer beats ask beats allow.
- **Use them for:** blocking, scoping, redacting, auditing, notifying.

> ON THE EXAM
>
> Hooks run every time, whatever the model decides. That is the guide's "hooks for deterministic actions".

How a tool call is permitted

01
Hooks
Can deny, even in bypassPermissions mode.

›

02
Deny rules
disallowedTools: blocked outright.

›

03
Ask rules
Routed for a decision.

›

04
Mode
default, acceptEdits, plan, dontAsk, bypass…

›

05
Allow, then callback
allowedTools; finally canUseTool decides.

THE TIEBREAKER

allowedTools pre-approves; it does not restrict bypassPermissions. A locked-down agent is allowedTools plus dontAsk.

Sessions

- **Continue, resume or fork:** pick up the latest session, a session by ID, or branch from one.
- **Sessions persist the conversation, not the filesystem.**
- **Stored as JSONL** on the host by default; a SessionStore moves them between hosts.
- **Compaction** keeps long sessions inside the window automatically.
- **Ephemeral containers lose local sessions** unless you store them elsewhere.

> THE TIEBREAKER
>
> Resuming a session restores what was said, not what was on disk. Files need their own storage.

SDK, or your own loop?

THE AGENT SDK

Use the engine

- Agent loop, tools, compaction, sessions
- Hooks and permissions built in
- Maintained by Anthropic
- Fits most agents

YOUR OWN LOOP

Only with a reason

- Messages API plus the tool loop from module 20
- Or the client SDK's beta tool runner
- Justified by a specific need the SDK cannot meet
- Every customisation owned for its life

ON THE EXAM

Judge each existing customisation one by one: keep it only if it meets a need the platform does not.

Where agents run

OPTION	WHAT YOU GET	WATCH FOR
Self-hosted SDK	Your container, your network, your data path	Sandboxing, scaling, session storage
Managed Agents	Anthropic-hosted harness and sandbox, sessions, events	Beta; not ZDR or HIPAA eligible
Managed, self-hosted sandbox	Anthropic's harness, your sandbox	You run the environment

CURRENCY CHECK

Anthropic: "the SDK runs in a process you operate, while Managed Agents runs in Anthropic's infrastructure."

Self-hosting safely

- **A sandboxed container per agent session:** Docker, gVisor, Firecracker or a sandbox service.
- **Egress through a proxy:** only the hosts the agent needs.
- **Start at about 1 GiB RAM, 5 GiB disk and 1 CPU** per agent, then measure.
- **Multi-tenant:** no shared settings or memory; a separate config directory and working directory per tenant.
- **Secrets injected at run time,** never baked into images or prompts.

> THE TAKEAWAY
>
> An agent with Bash is a program that writes programs. Contain it like one.

Copperlet: deploying the agency-ops agent

CHOICE

Self-hosted Agent SDK

- Tenant data stays in Copperlet's cloud account
- Harwood cases must run through Bedrock
- Managed Agents not ZDR-eligible; DPO declined
- Already runs containers and a job queue

HOW IT RUNS

Contained and bounded

- One gVisor container per case, from the job queue
- Egress only to the MCP server and Claude endpoints
- Sessions in a SessionStore; containers ephemeral
- Hooks audit every MCP call

THE TAKEAWAY

Requirements chose the hosting, not fashion: data path, platform and the DPO's review.

> A team spent three months building its own agent loop with history management, tool dispatch, compaction and permission checks. It is buggy, and new team members struggle to maintain it. Nothing in their requirements is unusual.
>
> They are considering next steps.

What should they do?

A. Keep extending the custom loop, since so much work has gone into it

B. Move to the Agent SDK, keeping only customisations it lacks

C. Rewrite the custom loop from scratch with better tests

D. Adopt a second framework on top of the custom loop

◆ PAUSE AND DECIDE

What should they do?

A Keep extending the custom loop, since so much work has gone into it

Sunk cost; the bugs stay theirs.

B Move to the Agent SDK, keeping only customisations it lacks

Use the platform; judge each customisation.

C Rewrite the custom loop from scratch with better tests

Rebuilds what the SDK provides.

D Adopt a second framework on top of the custom loop

Two layers to maintain.

THE SIGNAL

"Nothing unusual" plus a home-made loop: use the SDK; keep only what it cannot do.

> An Agent SDK app must never run shell commands that delete files outside a working folder, whatever the model decides. Currently the rule is only in the system prompt, and testing shows rare violations.
>
> The team wants a guarantee.

What should they add?

A. Repeat the rule more strongly in the system prompt

B. Lower the agent's max_turns

C. Ask the model to confirm before every command

D. A PreToolUse hook on Bash that denies violations

◆ PAUSE AND DECIDE

What should they add?

A Repeat the rule more strongly in the system prompt

Still guidance.

B Lower the agent's max_turns

Fewer turns, same risk per turn.

C Ask the model to confirm before every command

The model checking itself.

D A PreToolUse hook on Bash that denies violations

Deterministic, every time.

THE SIGNAL

"Whatever the model decides" means a hook. A hook deny holds even in bypass mode.

> A reporting agent should use exactly three read-only tools and nothing else, with no interactive approval prompts, because it runs unattended overnight.
>
> Today it sometimes stalls waiting for an approval nobody gives.

How should permissions be configured?

A. Allow the three tools in allowedTools, with dontAsk mode

B. Use bypassPermissions, so nothing ever waits

C. Leave the default mode and tell the agent not to use other tools

D. Remove all hooks, so no prompts appear

◆ PAUSE AND DECIDE

How should permissions be configured?

A. Allow the three tools in allowedTools, with dontAsk mode

Exactly three tools, never a prompt.

B. Use bypassPermissions, so nothing ever waits

Allows every tool, not three.

C. Leave the default mode and tell the agent not to use other tools

Still stalls; still only a request.

D. Remove all hooks, so no prompts appear

Hooks are not what prompts.

CURRENCY CHECK

A locked-down, unattended agent: allowedTools plus dontAsk. Bypass allows everything.

> A company will run coding agents for many customers on its own infrastructure, using the Agent SDK. Agents can run Bash, and customer code must never mix. Containers are created per job and destroyed afterwards.
>
> The team wants a safe hosting design.

Which two practices fit best? Choose two.

- A Run each session in a sandboxed container, with egress only to approved hosts
- B Share one long-running container across customers to save cost
- C Isolate tenants, and keep sessions in an external store
- D Bake each customer's API keys into the container image
- E Rely on the system prompt to keep customers' files apart

◆ PAUSE AND DECIDE

Which two practices fit best? Choose two.

A Run each session in a sandboxed container, with egress only to approved hosts

Contains what Bash can do.

B Share one long-running container across customers to save cost

Exactly the mixing to avoid.

C Isolate tenants, and keep sessions in an external store

No leaks; sessions survive.

D Bake each customer's API keys into the container image

Secrets in images leak.

E Rely on the system prompt to keep customers' files apart

Guidance, not isolation.

THE SIGNAL

Bash plus many tenants: sandbox each session, isolate each tenant, store sessions off the container.

> A small start-up wants a long-running research agent with a sandbox, sessions and scheduling, and has no infrastructure team. Its data has no residency or zero-retention requirements.
>
> It wants to ship quickly.

Which option fits best?

A Build a custom container platform for the Agent SDK first

B Write its own agent loop on the Messages API

C Claude Managed Agents, Anthropic's hosted harness and sandbox

D Run the agent on a developer's laptop each night

◆ PAUSE AND DECIDE

Which option fits best?

A Build a custom container platform for the Agent SDK first

Infrastructure they cannot staff.

B Write its own agent loop on the Messages API

Rebuilds the harness, and still needs hosting.

C Claude Managed Agents, Anthropic's hosted harness and sandbox

Hosted, with no blocking requirement.

D Run the agent on a developer's laptop each night

Not a production design.

THE SIGNAL

No infrastructure team, no ZDR or residency requirement, ship fast: Anthropic-hosted fits.

> An agent runs in ephemeral containers. When a long job is interrupted and restarted in a new container, the agent has lost the conversation and starts the case again from scratch.
>
> The team expected resume to work.

What should they change?

A Increase the container's disk size

B Keep sessions in a SessionStore, and resume by ID

C Raise max_turns, so jobs finish before interruptions

D Turn off compaction, so nothing is summarised away

◆ PAUSE AND DECIDE

What should they change?

A Increase the container's disk size

The disk is discarded anyway.

B Keep sessions in a SessionStore, and resume by ID

Sessions outlive the container.

C Raise max_turns, so jobs finish before interruptions

Unrelated to interruptions.

D Turn off compaction, so nothing is summarised away

Wrong cause; risks overflow.

THE SIGNAL

Sessions are stored on the host by default. Ephemeral hosts need a SessionStore.

Module 25 in five lines

- **The Agent SDK** gives Claude Code's loop, tools, compaction, sessions, hooks and permissions.
- **Set limits yourself:** max turns and a spend budget default to none.
- **Hooks enforce;** permissions run hooks, deny, ask, mode, allow, then canUseTool.
- **Own loop only for a real gap;** judge customisations one by one.
- **Self-host in sandboxes,** or use Managed Agents when requirements allow.

COMING UP

Next: agent patterns, memory and frameworks.